

In[\*]:= Quit

In[\*]:=  $\epsilon = 1$

Out[\*]=

1

(\*\*Ejemplo 1

In[\*]:=  $P[t_, x_] = \frac{1}{1 + t x}$

Out[\*]=

$$\frac{1}{1 + t x}$$

In[\*]:= P[1, 1]

Out[\*]=

$$\frac{1}{2}$$

In[\*]:= P[1, 1 +  $\epsilon$ ] // N

Out[\*]=

0.333333

(\*\*Construyendo la aproximación

In[\*]:= DxP[t\_, x\_] = D[P[t, x], x]

Out[\*]=

$$-\frac{t}{(1 + t x)^2}$$

In[\*]:= DDxxP[t\_, x\_] = D[DxP[t, x], x, x]

Out[\*]=

$$\frac{2 t^2}{(1 + t x)^3}$$

In[\*]:= Aprox = P[1, 1] + DxP[1, 1]  $\epsilon$  +  $\frac{1}{2}$  DDxxP[1, 1]  $\epsilon^2$  // N

Out[\*]=

0.375

In[\*]:= Quit

In[\*]:=  $\epsilon = \frac{1}{1000}$

Out[\*]=

$$\frac{1}{1000}$$

(\*\*Ejemplo 2

$$\text{In[*]}:= \text{P}[\mathbf{t\_}, \mathbf{x\_}] = \frac{1}{1 + \text{Cos}[\mathbf{t}] \mathbf{x}^7}$$

$$\text{Out[*]}= \frac{1}{1 + \mathbf{x}^7 \text{Cos}[\mathbf{t}]}$$

$$\text{In[*]}:= \text{P}[\mathbf{1}, \mathbf{1}] // \mathbf{N}$$

$$\text{Out[*]}= 0.649223$$

$$\text{In[*]}:= \text{P}[\mathbf{1}, \mathbf{1} + \epsilon] // \mathbf{N}$$

$$\text{Out[*]}= 0.647628$$

**(\*\*Construyendo la aproximación**

$$\text{In[*]}:= \text{DxP}[\mathbf{t\_}, \mathbf{x\_}] = \text{D}[\text{P}[\mathbf{t}, \mathbf{x}], \mathbf{x}]$$

$$\text{Out[*]}= -\frac{7 \mathbf{x}^6 \text{Cos}[\mathbf{t}]}{\left(1 + \mathbf{x}^7 \text{Cos}[\mathbf{t}]\right)^2}$$

$$\text{In[*]}:= \text{DDxxP}[\mathbf{t\_}, \mathbf{x\_}] = \text{D}[\text{P}[\mathbf{t}, \mathbf{x}], \mathbf{x}, \mathbf{x}]$$

$$\text{Out[*]}= \frac{98 \mathbf{x}^{12} \text{Cos}[\mathbf{t}]^2}{\left(1 + \mathbf{x}^7 \text{Cos}[\mathbf{t}]\right)^3} - \frac{42 \mathbf{x}^5 \text{Cos}[\mathbf{t}]}{\left(1 + \mathbf{x}^7 \text{Cos}[\mathbf{t}]\right)^2}$$

$$\text{In[*]}:= \text{Aprox} = \text{P}[\mathbf{1}, \mathbf{1}] + \text{DxP}[\mathbf{1}, \mathbf{1}] \epsilon + \frac{1}{2} \text{DDxxP}[\mathbf{1}, \mathbf{1}] \epsilon^2 // \mathbf{N}$$

$$\text{Out[*]}= 0.647628$$